அத்தியாயம் 10

பயிற்சி 10.1

1. (i) 1,1 (ii) 3,2 (iii) 2, இருத்தல் அல்ல (iv) 1, 2 (v) 1,4
   (vi) 2,2 (vii) 2,6 (viii) 2, இருத்தல் அல்ல (ix) 3,1 (x) 1, 1

பயிற்சி 10.2

1. (i) $\frac{dQ}{dt} = kQ$ (ii) $\frac{dP}{dt} = kP(500000 - P)$ (iii) $\frac{dP}{dT} = \frac{kP}{T^2}$ (iv) $\frac{dx}{dt} = \frac{2x}{25} + 400$ 2. $\frac{dr}{dt} = -k$

பயிற்சி 10.3

1. (i) $\frac{d^2y}{dx^2} = 0$ (ii) $\frac{d^2x}{dy^2} = 0$ 2. $r^2\left[1+\left(\frac{dy}{dx}\right)^2\right] = \left(x\frac{dy}{dx} - y\right)^2$

3. $x^2 + 2xy\frac{dy}{dx} - y^2 = 0$ 4. $2ay'' + y'^3 = 0$ 5. $xy' - 2y - 2 = 0$ 6. $xy'^2 + xyy'' - yy' = 0$

7. $\frac{d^2y}{dx^2} = 64y$ 8. $xy'' + 2y' + x^2 - xy - 2 = 0$

பயிற்சி 10.4

2. (i) $m = -2$ (ii) $m = 2,3$ 3. $2y^2 = x + 48$

பயிற்சி 10.5

1. $F = (F - kV)e^{\frac{kt}{M}}$ 2. $k^2\left(1 - e^{-\frac{2gx}{k^2}}\right) = v^2$ 3. $y = \frac{1-x}{1+x}$

4. (i) $\sin^{-1} y = \sin^{-1} x + C$ (ii) $y\tan^{-1} x = C$ (iii) $\sin\left(\frac{y-1}{x}\right) = a$ (iv) $e^x + e^{-y} + \frac{x^4}{4} = C$
   (v) $\left(e^y + 1\right)\sin x = C$ (vi) $\sin\left(\frac{x}{y}\right) = e^{nx+c}$ (vii) $3y = -\left(25 - x^2\right)^{\frac{3}{2}} + 3C$ (viii) $\sin y = e^x \log x + C$
   (ix) $\sec y = 2\sin x + C$ (x) $\frac{1}{2}\left[(x+y) + \sin(x+y)\cos(x+y)\right] = x + C$

பயிற்சி 10.6

1. $\sin\left(\frac{y}{x}\right) = \log|Cx|$ 2. $y = Ce^{\frac{x^3}{3y^3}}$ 3. $e^{\frac{x}{y}} = \log|Cy|$ 4. $3x^2y + 2y^3 = C$
5. $xy^2 - x^2y = C$ 6. $C = xe^{\tan\left(\frac{y}{x}\right)}$ 7. $y + 3xe^{\frac{y}{x}} = C$ 8. $x_0 = \pm\sqrt{3}e$

பயிற்சி 10.7

1. $y = \sin x + C\cos x$ 2. $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}} + C\left(1-x^2\right)^{-\frac{1}{2}}$ 3. $(y + \cos x)x = \sin x + C$

4. $y\left(x^2+1\right) = \frac{x}{2}\sqrt{x^2+4} + \frac{1}{2}\log\left|x + \sqrt{x^2+4}\right| + C$ 5. $xy^2 = 2y^5 + C$ 6. $xy\sin x + \cos x = C$

7. $ye^{\sin^{-1} x} = \frac{e^{2\sin^{-1} x}}{2} + C$ 8. $y\left(\frac{1+\sqrt{x}}{1-\sqrt{x}}\right) = x + \frac{2}{3}x\sqrt{x} + C$ 9. $xy + \tan^{-1} y = C$

10. $y\log x + \frac{\cos 2x}{2} = C$ 11. $2y = (x+a)^4 + 2C(x+a)^2$ 12. $y\left(1+x^3\right) = \frac{x}{2} - \frac{\sin 2x}{4} + C$

13. $4yx = 2x^2\log x - x^2 + 4C$ 14. $x^2y = \frac{x^4}{4}\log x - \frac{x^4}{16} + C$ 15. $2x^3y = x^2 + 3$

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